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Performance-limiting electron and hole traps in 4H-SiC PiN power diodes determined by combined DLTFS and TCAD studies

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ABSTRACT

The performance-limiting electron and hole trapping centers in 4H-SiC PiN power diodes are determined by combined deep-level transient Fourier spectroscopy (DLTFS) experiments and technology computer-aided design (TCAD) simulations. Two electron traps E_1 ($E_C - 0.19$ eV) and E_2 ($E_C - 0.67$ eV) and three hole traps H_1 ($E_V + 0.16$ eV), H_2 ($E_V + 0.3$ eV), and H_3 ($E_V + 0.63$ eV) are detected by DLTFS. Since DLTFS measurements were limited to 400 K, a few deep-level defects could not be detected in our experiments. In addition to the traps identified by DLTFS, two deep levels commonly reported at elevated temperatures, E_3 ($E_C - 1.65$ eV) and H_4 ($E_V + 1.43$ eV), are integrated into the TCAD model to perform a reliable trapping analysis. The effects of electron traps, hole traps, and individual traps are evaluated by selectively excluding them from the simulation. Hole trapping is found to be more prominent than electron trapping in pristine (as-fabricated/untouched) diodes. Among the traps, shallow hole trap H_1 exhibits a strong impact in reducing the conduction current (followed by E_3) in pristine diodes. To explore the fundamental nature of each trap, the concentration (N_T) of an individual trap is increased to a higher value without changing the N_T of other defects. Subsequently, the diode characteristics are analyzed at higher N_T of the specific trap. The traps E_2 and E_3 significantly reduce the diode current at higher N_T. The deep acceptor E_2 is primarily responsible for the donor doping compensation in the n^- drift layer.

I. INTRODUCTION

4H-silicon carbide (4H-SiC) PiN diodes are targeted for use in high-voltage energy transmission and smart-grid applications, as they exhibit high breakdown voltage ($V_{BR} > 10$ kV), low leakage current, and high surge current capability compared to the commercial 4H-SiC Schottky and junction barrier diodes (JBSs).[1,2] To achieve the desired off-state performance, p^+ junction termination extension (JTE) and p^+ floating guard rings are incorporated in the 4H-SiC PiN diode (p^+–n^-–n^+ structure) to suppress the peak electric field at the PN junction periphery.[1–4] The JTE and guard ring configurations are created in the 4H-SiC PiN diode by high-energy aluminum (Al^+) ion-implantation.[2,5,6] Relative to the unipolar 4H-SiC diodes (Schottky and JBS), the bipolar 4H-SiC PiN diode performance is affected by implantation-induced deep-level defects.[6–12] The carrier trapping effects undermine the diode rectification and conduction current, altering the epilayer doping density and switching properties. The trapping-induced degradations are governed by the defect parameters: energy level (E_T), trap concentration (N_T), capture cross section (σ), and trap charge state (donor or acceptor).[13–17]

The literature reports[10,15–17] have established the detrimental influence of deep-level electron traps ($Z_{1/2}$ and $EH_{6/7}$) on the forward current properties of the 4H-SiC PiN power diodes. Megherbi *et al.*[10,16] found that the defects $Z_{1/2}$ and $EH_{6/7}$ have a substantial impact on reducing forward current due to the trap-assisted electron–hole recombination mechanism. At the same time, the electron trapping effects on the reverse recovery

(RR) switching characteristics of the SiC PiN diode have not yet been analyzed; this study is essential as the PiN diode rectification critically depends on RR performance.[1] Note that the hole traps (located at $E_V + E_T$) were also identified in the 4H-SiC PiN diodes, in addition to the electron traps (placed at $E_C - E_T$).[9,11] The electron trapping effects in the SiC PiN diodes are primarily focused on the literature;[10,15–17] however, the detrimental effects induced by the hole traps on the 4H-SiC PiN diode properties remain unexplored. Specifically, the hole trapping-induced deviations in the *I–V, C–V*, and RR performance of the 4H-SiC PiN diode are not yet reported. Particularly, the individual impact of specific electron and hole traps on the static and transient current characteristics are not explicitly investigated. The present study addresses these research gaps through the combined deep-level transient Fourier spectroscopy (DLTFS) and technology computer-aided design (TCAD) approach.

In this work, the electron and hole trap parameters (E_T, N_T, and σ) in the PiN diode are characterized by the DLTFS experiments.[18–20] Although the DLTFS experiment determines the trap signatures in 4H-SiC diodes, TCAD simulations are necessary to evaluate the trapping-induced variations in the electrical and switching performance of the PiN diode. As a first in simulation, the TCAD physical model is precisely validated using the measured *I–V, C–V*, and RR data. Subsequently, the traps are selectively removed from the calibrated TCAD model to isolate the effects of electron trapping, hole trapping, and individual trapping. Accordingly, performance-limiting electron and hole traps in the pristine 4H-SiC PiN diodes are reported. To understand the fundamental physical nature of each trap, the diode properties are compared at a higher N_T of the particular trap, without changing the N_T of other defects. It is also essential to inspect the diode performance at higher N_T values, as the trap density typically increases when the diode is exposed to high-energy radiation.[14,21–23] The 4H-SiC diodes can be used in high-radiation environments, so the high-N_T effect study helps develop a TCAD-based radiation model for the 4H-SiC PiN diodes.[21,22] Finally, the performance-affecting defect states in the 4H-SiC PiN power diodes are determined in the pristine and higher N_T conditions.

II. EXPERIMENT

A. 4H-SiC PIN diode details

The 4H-SiC PiN diode fabrication and characterization procedures were described elsewhere.[2,24] The diode cross-sectional diagram representing the doping concentration and thickness of each layer is shown in Fig. 1. The p^+–n^-–n^+ structure was created in the wafer stack itself by the epilayer deposition (not by ion-implantation). After mesa isolation, the p^+ JTE and the p^+ floating guard rings were created by the Al^+ ion-implantation.[2] The active area of the PiN diode used for this work is about 59 mm^2.

B. DLTFS measurement

The DLTFS characterization setup and procedure were discussed in our earlier reports.[20,25] To determine the physical location of the defect states, two different voltage pulsing schemes were employed: (a) majority carrier pulse and (b) carrier injection pulse.[13] In the majority carrier pulse, the diode voltage was switched from a reverse bias $V_R = -10$ V to a near-zero bias voltage $V_P = -0.1$ V for a voltage pulse width $t_P = 100\,\mu$s to populate majority carrier traps in the respective n-type and p-type layers of the PiN diode. In the carrier injection case, the diode bias voltage was pulsed from $V_R = -10$ V to a high-injection bias condition $V_P = +5$ V for $t_P = 100\,\mu$s to promote hole injection into the n^- drift layer; thus, the hole traps in the n-type layer may be filled under this condition. Subsequently, the bias voltage was resumed to V_R, and the capacitance emission transient signal was measured at two distinct period widths $T_W = 20.48$ ms and 2.048 s. The resultant DLTFS signal was denoted by the first-order Fourier sine coefficient b1.[18–20,25] The emission time constant (τ_e) of the trap was computed at regular temperature intervals by DLTFS software,[19] so a single DLTFS temperature scan may be sufficient to compute the trap signatures, such as trap activation energy and capture cross section.

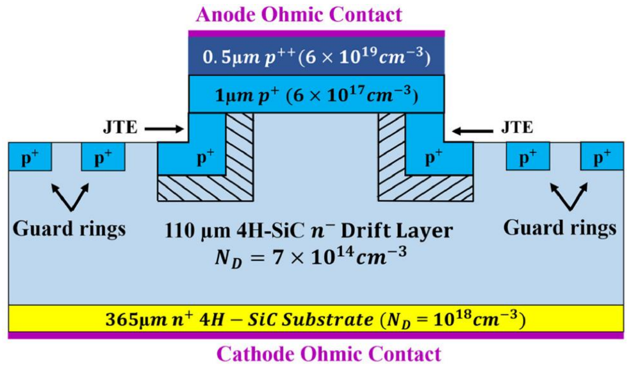

FIG. 1. The 4H-SiC PiN diode cross-sectional diagram, indicating the doping concentration and thickness of each layer in the wafer stack.

III. TCAD SIMULATION DETAILS

The 4H-SiC PiN diode structure in Fig. 1 is created in Sentaurus TCAD,[26] featuring a pure anode Ohmic contact and a resistive cathode Ohmic contact, with fine meshing applied at the junctions. The key physical models[26] are enabled in simulation, including bandgap narrowing, Fermi–Dirac statistics, incomplete ionization of acceptor dopants, temperature- and doping-dependent carrier mobility, trap-assisted SRH recombination, velocity saturation under high fields, Auger indirect recombination, and anisotropic mobility effects to capture the directional transport behavior of the hexagonal SiC lattice. The electron and hole trap signatures determined from the DLTFS measurements are incorporated in the TCAD physics section, along with the two more deep-level states at $E_C - 1.65$ eV and $E_V + 1.43$ eV (discussed in Sec. IV A). The trap-assisted SRH recombination statistics are modeled in the simulation using the following expression:[26]

$$R_{\rm net}^{SRH} = \frac{np - \gamma_n\gamma_p n_{i,\rm eff}^2}{\tau_p(n+\gamma_n n_1) + \tau_n(p+\gamma_p p_1)}, \tag{1}$$

where R_{net}^{SRH} is the SRH recombination rate, n is the free electron concentration in E_C, p is the free hole concentration in E_V, $n_{i,\mathrm{eff}}$ is the effective intrinsic concentration, τ_p is the hole lifetime, τ_n is the electron lifetime, n_1 and p_1 are the SRH model parameters in the Sentaurus TCAD,[26] and γ_n and γ_p are the correction factors required in the SRH model at higher doping levels. The doping dependence of the carrier lifetime (τ_{dop}) is incorporated into the SRH TCAD model,[26]

$$\tau_{\mathrm{dop}} = \tau_{\min} + \frac{\tau_{\max} - \tau_{\min}}{1 + \left(\frac{N_{A,0} + N_{D,0}}{N_{\mathrm{ref}}}\right)^{\gamma}}, \tag{2}$$

where N_A is the acceptor doping concentration in the p-layer, N_D is donor doping concentration in the n-layer, N_{ref} is the reference doping density, $\tau_{\min}$ is the minimum carrier lifetime, $\tau_{\max}$ is the maximum carrier lifetime, and γ is the fitting parameter. The trap-assisted SRH recombination model parameters $\tau_{\max} = 2.5\,\mu$s for electron, $\tau_{\max} = 0.5\,\mu$s for hole, $\tau_{\min} = 0$ s for both electron and hole, $N_{\mathrm{ref}} = 3 \times 10^{17}\,\mathrm{cm}^{-3}$, and $\gamma_n = \gamma_p = 0.3$, are employed in the Sentaurus TCAD to match the measured electrical characteristics.

Surface recombination at the mesa sidewall periphery can reduce the free carrier density (through carrier recombination) during forward bias operation.[27] The recombination current modifies the diode's ideality factor ($n \approx 2$) in the low forward-voltage region and increases the on-state voltage drop,[1,27,28] thereby affecting the forward I–V characteristics. The 4H-SiC PiN diode used in this study has JTE periphery protection along the mesa sidewalls (in n^- drift region). The minority carrier injection process occurs at a sufficiently larger distance from the mesa sidewalls due to the JTE structure so that the surface recombination effects may be less pronounced in the JTE-integrated PiN diode structure. Moreover, the typical surface recombination velocity in the JTE-integrated 4H-SiC PiN diode is unknown. Considering these points, the surface recombination effects were not explicitly included in the TCAD simulation model.

It is worth recalling that the PN junction structure was created in the wafer stack by high-quality epilayer deposition (not by ion-implantation). Hence, the interface states at the PN junction interface are expected to be minimal in the PiN diode. In literature reports,[7–12,29–32] the interface traps were not identified in the 4H-SiC PiN diodes (while the bulk traps were commonly detected), indicating that the interface traps may not be present in the 4H-SiC PiN diodes. The interface trapping effects are crucial at the SiO_2/4H-SiC interface regions of MOSFETs and MOS-based devices;[6,27,33] such a SiO_2/4H-SiC interface does not exist in the 4H-SiC PiN diode. Hence, the interface traps are not included in the PiN diode simulation model. However, the interface trap states and surface recombination may become increasingly important under extreme operating conditions, such as elevated temperatures, high surface electric fields, or devices with significant interface-controlled transport (for example, MOSFETs and MOS-based devices).

After incorporating all the relevant physical models and trap parameters, the forward I_F–V_F characteristics are simulated. For the C–V simulation, a small AC signal (30 mV) is superimposed at 1 MHz frequency during the reverse DC bias voltage sweep (0 to −20 V).[21] The diode depletion capacitance is then extracted as a function of the reverse voltage (V_R). The RR characteristics are simulated using the integrated device and circuit TCAD simulation network. Following the TCAD RR testbench configuration reported in the literature,[34–37] the 4H-SiC PiN diode is connected in series with a 1 mΩ resistor and a 0.4 μH inductor to evaluate its switching performance. In the beginning, the PiN diode is forward biased at $V_F = +5$ V to ensure the conductivity modulation in the n^- drift layer, and then it is instantly switched to 2 kV ($V_R = -2000$ V) to trigger the RR phenomenon. Subsequently, the reverse recovery transient current is acquired over the μs time scale to inspect the turn-off dynamics of the diode. The reverse recovery time (T_{rr}) is determined using the standard definition, i.e., the time elapsed from the onset of the reverse recovery current until it returns to a negligible value after the recovery process. The reverse recovery charge (Q_{rr}) is computed by integrating the transient reverse recovery current over the reverse recovery interval (T_{rr}), corresponding to the area under the reverse recovery current vs time curve. Thus, Q_{rr} represents the total charge extracted from the PiN diode as it transitions from the forward-conducting state to the reverse-blocking state. The peak reverse recovery current (I_{prr}) is extracted from the maximum magnitude of the reverse current during the recovery transient.

The time-dependent carrier capture and emission dynamics are parameterized in the reverse recovery transient simulation by the equation[26,38]

$$\frac{\partial f_n}{\partial t} = \sum\nolimits_i (1 - f_n) c_i - f_n\, e_i, \tag{3}$$

where f_n and $1 - f_n$ denote the probability of the trap being occupied or unoccupied with an electron and c_i and e_i are the carrier capture and emission rate associated with the trap level. The electron capture rate ($c_{i,n}$) and hole capture rate ($c_{i,p}$) are symbolized by[26,38]

$$c_{i,n} = n\sigma v_{th}, \;\; c_{i,p} = p\sigma v_{th}\;, \tag{4}$$

where v_{th} represents the carrier thermal velocity. The $c_{i,n}$ and $c_{i,p}$ are determined by free electron and hole concentrations in the respective energy bands, as represented in Eq. (4). The emission rate (e_i) of a carrier from a trap level is described by the following expression:[26,38]

$$e_i = \frac{\sigma\, v_{th}\, N_{C/V}}{g} \exp\left(\frac{-\Delta E_T}{kT}\right). \tag{5}$$

Here, $\Delta E_T = E_C - E_T$ is for the electron trap, while $\Delta E_T = E_T - E_V$ is for the hole trap, $N_{C/V}$ is the effective density of states in E_C (for electron trap) and E_V (for hole trap), g is the degeneracy factor, and T is the temperature. Equation (5) defines that the emission rate increases (emission time reduces) with increasing σ and decreasing ΔE_T. After matching the simulation results with the measured characteristics, the traps are selectively removed from the TCAD physical model to isolate the electron trapping and hole trapping and to inspect the influence of each trap state. To understand the fundamental trapping kinetics of each trap, the N_T of a

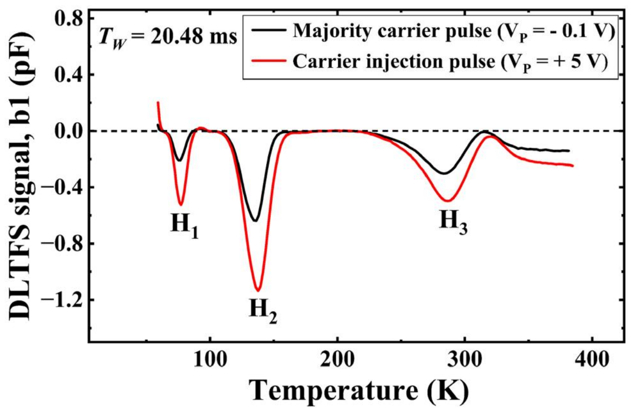

FIG. 2. DLTFS spectra measured at shorter $T_W = 20.48$ ms with the majority carrier pulse ($V_P = -0.1$ V) and carrier injection pulse ($V_P = +5$ V) conditions.

particular trap is increased without changing other trap parameters; the corresponding variations in the PiN diode characteristics are then evaluated. The I_F–V_F, C–V, and reverse recovery simulations were carried out at 300 K because the static electrical and RR characteristics of the 4H-SiC PiN diode were measured at room temperature (∼300 K).

IV. RESULTS AND DISCUSSION

A. DLTFS results

Figures 2 and 3 show the measured DLTFS at $T_W = 20.48$ ms and 2.048 s, upon the majority carrier pulse and carrier injection pulse conditions. The negative (H_1, H_2, and H_3) and positive (E_1 and E_2) peaks in our DLTFS experimental setup[20,25] indicate hole traps (at $E_V + E_T$) and electron traps (at $E_C - E_T$), respectively. An apparent increase in the DLTFS signal amplitude of the hole traps

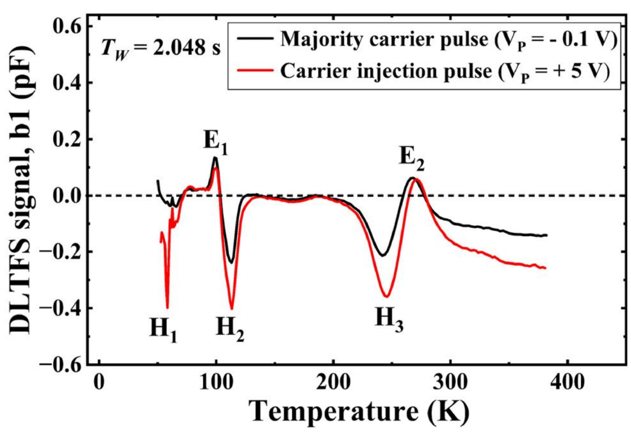

FIG. 3. DLTFS spectra acquired at longer $T_W = 2.048$ s upon the majority carrier pulse ($V_P = -0.1$ V) and carrier injection pulse ($V_P = +5$ V).

is observed in both Figs. 2 and 3, for the carrier injection pulse case ($V_R = -10$ V to $V_P = +5$ V), whereas the electron traps do not follow this trend. Note that the number of hole traps filled in the diode depends on the filling-pulse bias condition. During the majority carrier pulse (from $V_R = -10$ V to $V_P = -0.1$ V), no minority carrier injection (no hole injection) occurs in the n^- drift layer. In this bias condition, the hole traps in the n^- drift layer are not filled by the majority carrier pulse (only hole traps in the p^+ layer are populated).[13] Whereas the carrier injection pulse (from $V_R = -10$ V to $V_P = +5$ V) introduces a significant number of holes into the n^- drift layer;[13] hence, the hole traps in the n^- drift layer are also filled by this pulse. The filled traps only contribute to the DLTFS signal during the temperature scan. Since a relatively higher number of hole traps are populated during the carrier injection pulse (compared to the majority carrier pulse), an increase in the DLTFS signal amplitude is noticed for the hole traps in Figs. 2 and 3.

Figure 4 illustrates the $\ln(\tau_e T^2)$ vs $1/kT$ Arrhenius plots for the traps H_1, H_2, H_3, E_1, and E_2, constructed from the DLTFS spectra measured at $T_W = 2.048$ s with the carrier injection pulse. The Arrhenius plots of all traps yield a linear fit, indicating that the carrier emission rate follows the standard temperature dependence described by the Arrhenius law. The E_T and σ of the defect states are calculated from the slope and y-intercept of the DLTFS Arrhenius plots according to the equation[19,20,25]

$$\ln(\tau_e T^2) = \frac{E_T}{kT} - \ln\left(\frac{\sigma_n N_C v_{th}}{gT^2}\right). \tag{6}$$

Note that the transverse and longitudinal electron effective mass of 4H-SiC is 0.42 m_0 and 0.33 m_0, respectively (m_0 is the free

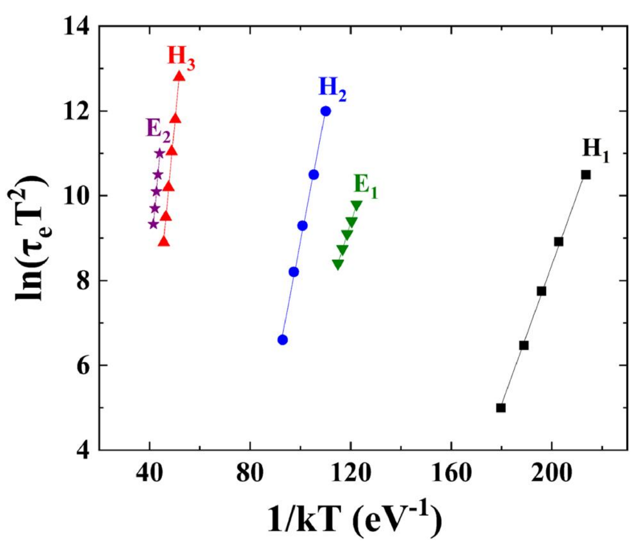

FIG. 4. $\ln(\tau_e T^2)$ vs $1/kT$ Arrhenius plots for the traps H_1, H_2, H_3, E_1, and E_2, constructed from the DLTFS spectra measured at $T_W = 2.048$ s with the carrier injection pulse.

TABLE I. The electron and hole trap parameters deduced by the DLTFS measurement and two more commonly reported deep-level traps in the 4H-SiC PiN diodes.

Trap label	Trap type	E_T	σ	N_T	Physical origin
E_1	Acceptor	$E_C - 0.19$ eV	3×10^{-18} to 5×10^{-18} cm^2	~10^{12} cm^{-3}	Ti
E_2	Acceptor	$E_C - 0.67$ eV	6×10^{-15} to 6.2×10^{-15} cm^2	~2.5×10^{11} cm^{-3}	$Z_{1/2}$
H_1	Acceptor	$E_V + 0.16$ eV	5.3×10^{-13} to 5.7×10^{-13} cm^2	(3–4) $\times 10^{12}$ cm^{-3}	Al, L-center
H_2	Acceptor	$E_V + 0.3$ eV	9×10^{-15} to 1×10^{-14} cm^2	(4–9) $\times 10^{12}$ cm^{-3}	Al, B, UK1
H_3	Acceptor	$E_V + 0.63$ eV	2×10^{-14} to 2.2×10^{-14} cm^2	(3–4) $\times 10^{12}$ cm^{-3}	D-center, HK1
Two more traps frequently detected in the 4H-SiC PiN diodes in literature[6,9,30–33]					
E_3	Donor	$E_C - 1.65$ eV	~2×10^{-14} cm^2	~2.5×10^{11} cm^{-3}	$EH_{6/7}$
H_4	Acceptor	$E_V + 1.43$ eV	~7×10^{-13} cm^2	~2×10^{12} cm^{-3}	HK4

electron mass).[6] The density of states (DOS) effective mass of the electron (m_n^*) is 0.8 m_0 in the 4H-SiC. The hole effective mass for the transverse and longitudinal components is 0.66 m_0 and 1.75 m_0. The DOS effective mass of hole (m_p^*) is 0.91 m_0 in the 4H-SiC.[6] The DOS effective mass of the electron (m_n^*) and hole (m_p^*) is used to compute the effective density states in the conduction and valence bands (N_C and N_V) and the carrier thermal velocity (v_{th}) parameters in Eq. (6). The N_T of the trap is computed from the DLTFS signal amplitude (b1 peak magnitude) as per the following expression:[19,20,25]

$$N_T \approx 2N_D \frac{\Delta C}{C_R}, \tag{7}$$

where N_D represents the doping concentration of the n$^-$ 4H-SiC drift layer, ΔC is the DLTFS peak amplitude, and C_R is the diode depletion capacitance at $V_R = -10$ V. The electron and hole trap signatures deduced from the DLTFS measurement are compiled in Table I. Note that the electron traps E_1 and E_2 are observed only at longer $T_W = 2.048$ s. The N_T of the electron traps E_1 and E_2 is significantly lower than that of the hole traps, and they may require a longer emission time window for electron detrapping. At $T_W = 2.048$ s, a sufficiently extended time is available to promote the electron emission from the traps E_1 and E_2. Possibly due to this reason, the low N_T electron traps are identified at longer $T_W = 2.048$ s.

The electron traps $E_C - 0.19$ eV (E_1) and $E_C - 0.67$ eV (E_2) are usually observed in n-type 4H-SiC epilayers[6,14,39–41] and are ascribed to the Ti impurity and the carbon-vacancy-related acceptor defect $Z_{1/2}$. Here, E_1 and E_2 are omnipresent trap levels in the n$^-$ 4H-SiC drift layer.[6,14,21,39–42] The hole trap $E_V + 0.16$ eV (H_1) may originate from Al-impurity acceptor states and L-centers.[43] The second hole trap $E_V + 0.3$ eV (H_2) may be associated with Al-related or B-related acceptor impurities[7,43] and the UK1 defect.[44] The possible origin of the defect $E_V + 0.63$ eV (H_3) can be the acceptor-like D-center[6,43] and HK1 defect.[44]

Note that an electron trap E_3 ($EH_{6/7}$) and a hole trap H_4 (summarized in Table I) were not identified in the current DLTFS study due to the cryostat's temperature limit (because these deep-level defects are typically detected at higher temperatures $T > 500$ K). The carbon-vacancy-related donor-type trap E_3 at $E_C - 1.65$ eV is widely reported in n-type 4H-SiC layers[6,39–42] and is also considered an omnipresent defect in the n$^-$ drift region (like E_1 and E_2). Since the trap E_3 ($EH_{6/7}$) is positioned near the middle of the bandgap, it effectively acts as a generation–recombination center,[6] thereby limiting the minority-carrier injection process in the PiN diode. If the $EH_{6/7}$ trap is not incorporated in the model, the carrier generation–recombination effects may be incorrectly simulated. Additionally, in our earlier work,[9] a deep hole trap $E_V + 1.43$ eV (H_4) was also identified in the SiC PiN diodes fabricated on the same wafer set. This trap H_4 is attributed to the HK4 center in the p-type 4H-SiC epilayer.[44] Hence, both traps E_3 and H_4 were incorporated into the TCAD simulation to accurately capture the realistic electrical characteristics and trapping effects in the 4H-SiC PiN diode. If the traps E_3 ($EH_{6/7}$) and H_4 are not included in the simulation model, the trapping effects on the PiN diode characteristics could be inaccurately anticipated because these traps are inherently present in the 4H-SiC PiN diodes. Among these traps, only the trap E_3 is found as a donor-like state;[6,21,22,45] all remaining defects are considered as acceptor-type traps in the TCAD model, based on the literature reports.[6–17,21,22,39–45]

To perform reliable trapping analysis, the defects (E_1, E_2, H_1, H_2, and H_3) identified from the DLTFS are included in the TCAD model along with the deep-level defects E_3 and H_4. The electron traps E_1, E_2, and E_3 are placed in the n$^-$ 4H-SiC drift layer in the simulation, as they are omnipresent in the n-type 4H-SiC.[6,14,39–42] While the spatial location of the hole traps is probed by analyzing the trap-filling bias conditions. During the majority carrier pulse, electron traps in the n$^-$ 4H-SiC drift layer and hole traps in the p-type 4H-SiC layer are filled,[13,20,46,47] as shown in Fig. 5. A slight change in the p$^+$-side depletion thickness is anticipated when the reverse voltage is reduced from $V_R = -10$ V to $V_P = -0.1$ V due to the high acceptor doping density in the p$^+$ layer ($N_A = 6 \times 10^{17}$ cm^{-3}) compared to the drift layer N_D (7×10^{14} cm^{-3}). Hence, hole trap filling in the p$^+$ layer due to the majority carrier pulse is expected to be small. The depletion region in the lightly doped n$^-$ layer should rapidly decrease during the majority carrier pulse, thus promoting the electron trap filling in that region.[13,20,46,47] However, the N_T of hole traps is found to be higher than that of electron traps even for the majority carrier pulse. This observation suggests that the hole traps are in the p$^+$ epilayer and p$^+$ JTE of the PiN diode. In the literature,[6,7,44] the hole traps were typically detected in the p-type 4H-SiC epitaxial layers, as supporting our hypothesis. Intriguingly, the N_T of hole traps is further increased when the PiN diode is exposed to the

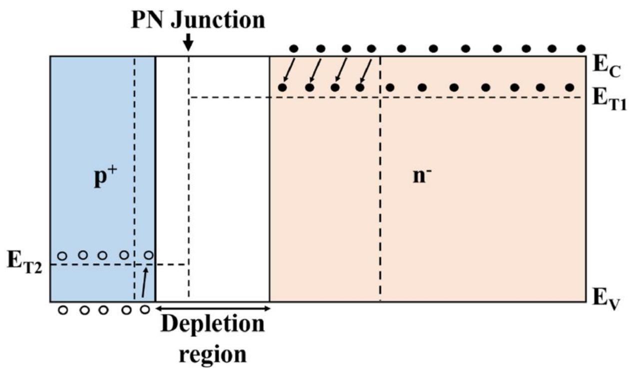

FIG. 5. Majority carrier traps in the respective n-type and p-type 4H-SiC layers filled when the voltage pulsed from $V_R = -10$ V to $V_P = -0.1$ V (under majority carrier pulse).

carrier injection pulse. At a high-injection condition ($V_P = +5$ V), the injected hole concentration in the n^- drift layer should exceed the background donor concentration, which modulates the drift layer conductivity.[1,12,48] Under this bias condition, the hole traps in the n^- drift layer may be populated by the injected holes,[13,46–48] as depicted in Fig. 6. Nonetheless, the hole traps H_1–H_4 were rarely observed in the n-type 4H-SiC Schottky and JBS diodes. Accordingly, it is considered that the Al^+-implantation induces the lattice crystal defects in the p^+ JTE as well as the n^- drift layer located close to the JTE region (hatched portions in Fig. 1). Based on these inspections, in the TCAD simulation, the hole traps H_1–H_4 are introduced in the p^+ layer, p^+ JTE, and hatched portions (0.5 μm distance from the JTE region) in the n^- drift layer.

B. TCAD model validation

Since the carrier injection voltage pulse fills more traps in the 4H-SiC PiN diodes, they should reflect the true N_T value of those traps. Accordingly, the highest N_T value identified for each trap

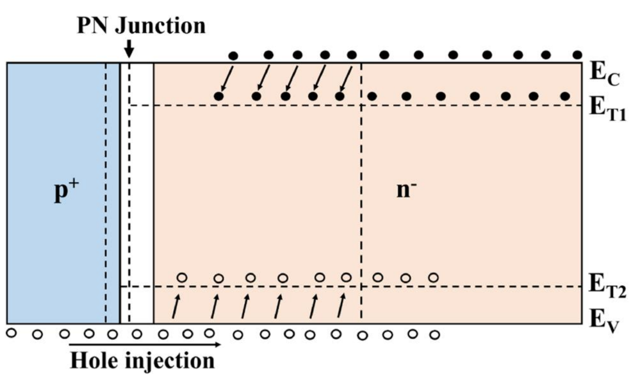

FIG. 6. Trapping mechanism in the 4H-SiC PiN diode induced by the carrier injection pulse when the voltage switched from $V_R = -10$ V to $V_P = +5$ V (under carrier injection pulse).

(from the range of N_T values) in Table I is chosen in the simulation model. Similarly, the highest σ value of the trap in Table I is considered for the simulation analysis. Figures 7(a) and 7(b) show an excellent agreement between the measured and simulated I_F–V_F characteristics in both linear and semi-log scales. Figure 7(a) indicates that the diode turn-on voltage ($V_{on} = 2.7$ V at 1 mA) predicted by the simulation closely tracks the experimental value of ∼2.65 V. Since multiple current transport mechanisms govern the forward conduction in a 4H-SiC PiN diode, the linear-scale plot in Fig. 7(a) does not clearly distinguish these mechanisms. This differentiation can be better understood using the ideality factor n. In general, an ideality factor of $n = 1$ indicates diffusion-dominated current due to low-level injection, whereas $n = 2$ suggests recombination-dominated current or the onset of high-level injection.[1] Therefore, the forward I_F–V_F characteristics are plotted on a semi-log scale in Fig. 7(b). It is observed that below 0.5 V, the current is dominated by noise and is therefore omitted. In the range $0.5\ \text{V} < V_F < 2.3\ \text{V}$, recombination current dominates, with $n \approx 1.95$ (closer to 2). This behavior arises from the electron–hole recombination in the p^+/n^- depletion region, as the carrier diffusion length may be smaller than the depletion depth at low forward voltages. For $2.3\ \text{V} < V_F < 2.7\ \text{V}$, minority carrier injection occurs, with hole diffusion from the p^+ 4H-SiC layer into the n^- 4H-SiC drift region and electrons diffusion into the p^+ region, resulting in low-level injection with $n \approx 1.05$. As V_F increases further, enhanced carrier injection leads to a transition toward high-level injection,[12,48,49] causing a noticeable bending of the I_F curve beyond approximately 2.7 V, with $n \approx 2.1$. The high-level carrier injection increases the minority carrier density well above the donor doping density in the n^- drift layer, thereby modulating the drift layer conductivity.[1,12,48,49]

At higher forward voltages, the effect of diode's series resistance ($Rs \approx 1.2\ \Omega$) becomes significant.[1,48] In this regime, the voltage drop across the PiN diode is increasingly dominated by the resistive components of the drift region, contacts, and bulk material, resulting in a deviation from the ideal exponential behavior. Consequently, the I_F–V_F curve exhibits a quasi-linear slope in the semi-log plot, indicating that the forward current transport is limited by resistive voltage drop within the PiN diode.

Figures 8(a) and 8(b) show the accurate validation of the simulated and measured C–V and $(1/C')^2$–V properties. Furthermore, the donor doping concentration (N_D) of the drift region, computed from the slope of the $(1/C')^2$–V_R plot,[21] yields a good agreement between experiment ($7.8 \times 10^{14}\ \text{cm}^{-3}$) and simulation ($7.6 \times 10^{14}\ \text{cm}^{-3}$) in Fig. 8(b). Similarly, the simulated RR transient characteristics nearly match the experimental data in Fig. 9, with a reverse recovery time (T_{rr}) of ∼75 ns, a reverse recovery charge (Q_{rr}) of 5.7 μC, and a peak reverse recovery current (I_{prr}) of −150 A. Note that the minority carriers stored in the n^- 4H-SiC drift region during forward bias operation induces the reverse recovery transient when the diode instantaneously switched to off-state (through the carrier removal process).[1] The turn-off transient current slope di/dt calculated from the simulation (10.2 A/ns) and experimental (10.4 A/ns) RR characteristics is also found to be closer, further demonstrating the validity of the transient simulation model. The di/dt is determined by several factors, including Q_{rr}, minority carrier lifetime, carrier recombination kinetics, reverse

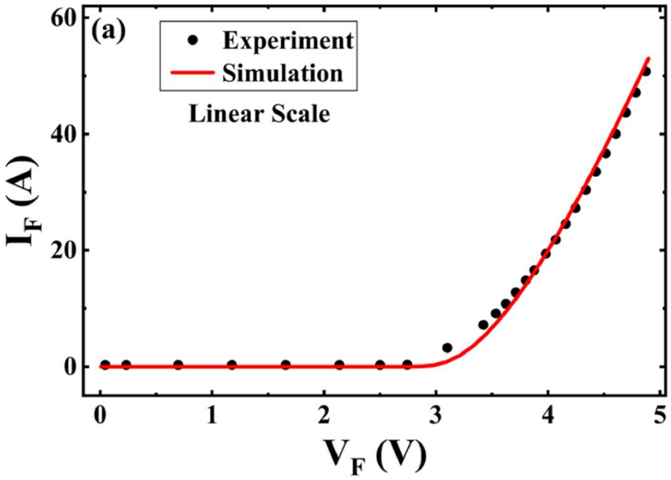

FIG. 7. Validation of simulated I_F–V_F plots of the 4H-SiC PiN power diode with the measurement I_F–V_F data in (a) linear and (b) semi-log scales.

bias voltage (V_R), circuit parasitic components, diode layer structure, and doping densities. Hence, the simulation model accurately captures the electrical and switching behavior of the PiN diode, making it a reliable tool for studying the trapping effects.

C. Isolating electron and hole trapping effects

It is worth noting that the validated simulation model comprises all seven trap levels mentioned in Table I. Nevertheless, it remains unknown which type of trap (electron or hole) has the most significant influence on the PiN diode characteristics. Therefore, the relative effects of the electron and hole traps are investigated using these four combinations: (i) including all traps, (ii) excluding hole traps only while keeping electron traps, (iii) excluding electron traps only while keeping hole traps, and (iv) excluding all traps.

Figure 10(a) illustrates the isolated electron and hole trapping effects on the I_F–V_F properties. There is no significant change in I_F for $V_F < 3$ V (below V_{ON}), and noticeable differences appear in the conductivity-modulated voltage range $V_F > 3$ V. So, the forward characteristics are shown only for higher $V_F > 3$ V in Fig. 10(a) to clearly illustrate the trapping-induced effects on the conduction properties. Figure 10(a) reveals that the inclusion of all traps in the simulation yields the lowest I_F, whereas the removal of all traps results in the highest I_F. This observation is consistent with expectations, as the absence of the carrier trapping facilitates unhindered forward current transport. Figure 11 compares the carrier concentration profile for two cases: (a) when all traps are included and (b) when all traps are removed. It is very clear that the injected electron and hole concentrations are higher for the case when all traps are removed, relative to the other case. The electron and hole densities

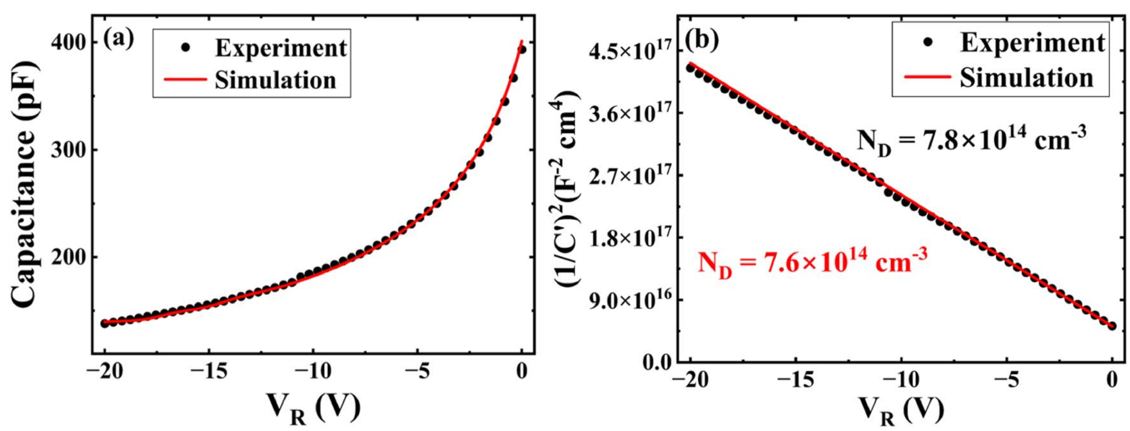

FIG. 8. Simulated (a) C–V and (b) $(1/C')^2$–V_R properties of the 4H-SiC PiN power diode are matched with the measured data.

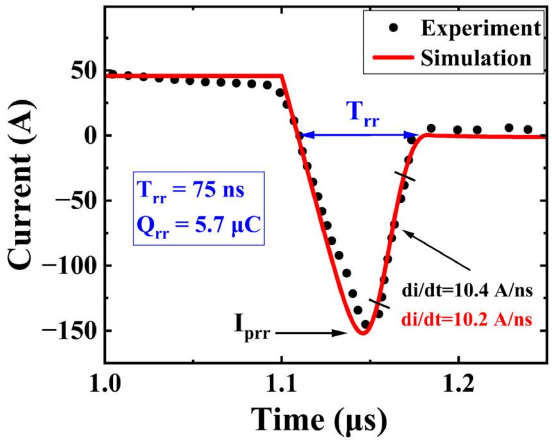

FIG. 9. Simulated reverse recovery (RR) transients of the 4H-SiC PiN power diode are matched with the measured RR data.

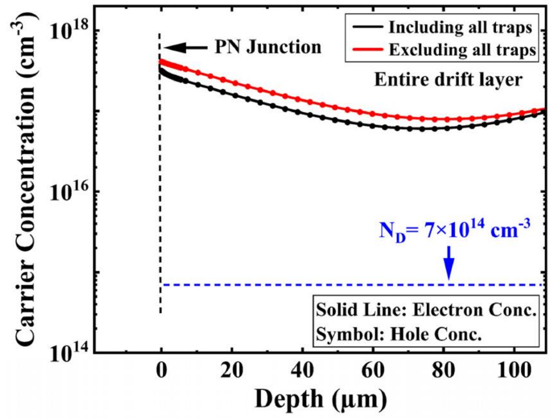

FIG. 11. The injected electron and hole concentration profiles in the entire drift layer at $V_F = 5$ V for the two extreme cases, such as including all traps and excluding all traps.

are nearly identical in the conductivity-modulated drift layer; for this reason, the two curves overlap each other in Fig. 11. It is also seen that the electron and hole concentrations surpass the background doping density N_D due to the high-level carrier injection. The absence of carrier trapping results in a substantially higher concentration of injected electrons and holes within the conductivity-modulated n$^-$ drift region. When the traps are removed, carrier capture is eliminated due to trapping, and a substantially larger concentration of injected electrons and holes is maintained in the conductivity-modulated drift region. In Fig. 10(a), removing all hole traps leads to a higher I_F than excluding electron traps. This behavior arises because the hole traps (H_1–H_4) typically possess larger N_T and σ values than the electron traps, as inferred from Table I.

The C–V characteristics are observed to be unaffected even after removing all traps in the model, as the N_T (10^{12} cm^{-3} range) of the traps is much smaller than the N_D (~7×10^{14} cm^{-3}) of the drift layer (not shown). The *RR* characteristics are plotted in Fig. 10(b) under identical conditions to inspect the electron and hole trapping influence on switching properties. Removing all traps yields the highest I_{prr} and longest T_{rr} due to the enhanced minority carrier storage, while including all traps gives the best RR performance. Similar to I_F–V_F, excluding hole traps from the model produces higher I_{prr} and longer T_{rr} than removing electron traps. When the hole traps are removed, fewer trapping centers remain (due to their higher N_T and σ), leading to an increased Q_{rr} that must be extracted during *RR*, which, in turn, raises I_{prr} and prolongs T_{rr}. Hence, removing hole traps deteriorates the *RR* behavior. The simulation results reveal that hole trapping severely impacts the electrical and switching behavior of the pristine PiN diodes (compared to electron trapping). Nevertheless, the

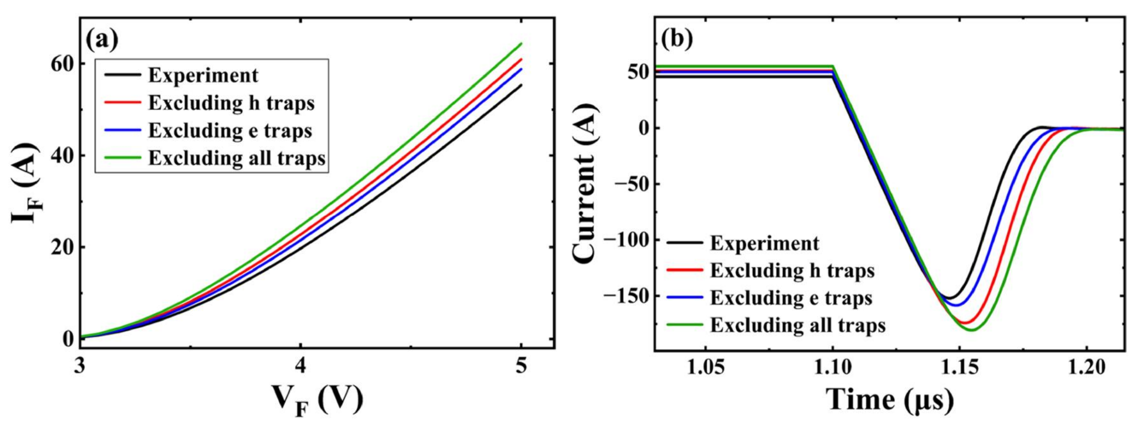

FIG. 10. Simulated (a) I_F–V_F and (b) *RR* properties for (i) including all traps, (ii) excluding hole (h) traps, (iii) excluding electron (e) traps, and (iv) no traps (excluding all traps).

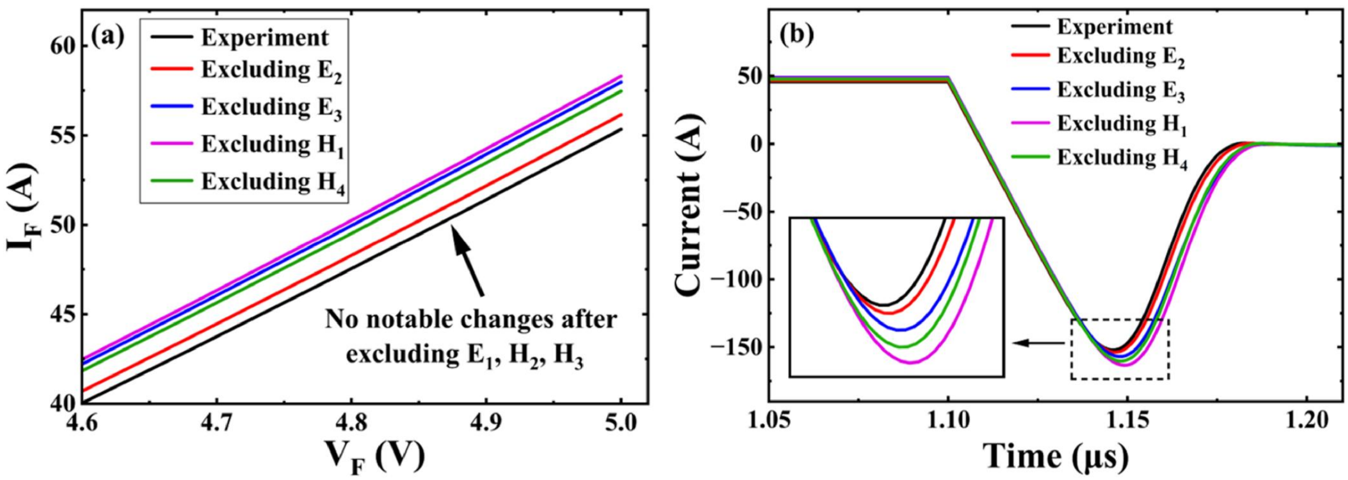

FIG. 12. (a) Simulation of I_F–V_F properties upon selectively excluding the defects E_2, E_3, H_1, and H_4 (higher V_F range >4.6 V is shown for clear visualization); (b) simulation of RR properties for the same trap removal conditions.

contribution of individual traps remains unclear, so the effect of each trap is separately investigated below in Sec. IV D.

D. Performance-limiting traps in pristine diodes

Figure 12(a) displays the simulated I_F–V_F characteristics by separately excluding the defects E_2, E_3, H_1, and H_4 from the simulation model. Figure 12(a) is shown at a high bias range ($V_F > 4.6$ V) for clear visualization of the results. Relative to the experimental I_F–V_F, a negligible deviation in I_F is noticed when the traps E_1, H_2, and H_3 are removed (not shown). Eliminating the shallow-level electron trap E_1 with a small σ (5×10^{-18} cm^2) barely changes I_F. Although H_2 and H_3 possess relatively higher N_T, their σ value is one order lower than that of other hole traps H_1 and H_4 (refer to Table I). For this reason, the hole traps H_2 and H_3 do not significantly affect the diode operation.

The hole trap $E_V + 0.16$ eV (H_1) is positioned close to the valence band compared to other hole traps; hence, it is ready to capture holes rapidly from the valence band.[13] Furthermore, the higher σ (5.7×10^{-13} cm^2) of H_1 enforces a fast hole capture process and reduces the free hole density during the forward conduction. Thus, excluding trap H_1 from the simulation allows unhindered carrier transport and results in the highest I_F. This sequence is followed by traps E_3, H_4, and E_2, as shown in Fig. 12(a). Trap E_3 is located near midgap and acts as an effective recombination center;[6,45] thus, its removal also enhances I_F. Similarly, eliminating the deep-level hole trap H_4 improves current due to its enhanced σ (7×10^{-13} cm^2) and recombination-assistive nature. The removal of deep-level E_2 results in a comparatively smaller yet still positive enhancement in forward current.

Figure 12(b) plots the simulated RR properties after excluding the defect states E_2, E_3, H_1, and H_4. Eliminating trap H_1 mitigates the carrier trapping, leading to higher I_{prr} and longer T_{rr}. A similar tendency (aligned with I_F–V_F) is noticed in the RR characteristics for the trap E_2. Unlike I_F–V_F properties, interchanging behavior is observed in Fig. 12(b) for E_3 and H_4, possibly due to the donor-like nature of trap E_3 and enhanced σ (7×10^{-13} cm^2) of the trap H_4. Table II summarizes the effects of each trap on the I_F and RR parameters (Q_{rr}, I_{prr}, and T_{rr}) of the diode. Accordingly, it is concluded that trap H_1 is the dominant detrimental defect in the pristine PiN diodes, followed by traps E_3 (although it has a midgap energy state), H_4, and E_2.

E. Trapping-effects at higher N_T

To understand the fundamental trapping dynamics of each defect in affecting electrical, switching, and carrier compensation (CC) properties, the diode characteristics are compared at higher N_T levels for each trap. This analysis is also useful to develop a radiation damage TCAD model for the irradiated 4H-SiC PiN diodes. For this purpose, N_T of a specific trap is selectively increased without changing other trap parameters (keeping their default values) in the TCAD physical model.

Figure 13(a) shows the simulated I_F–V_F plots for the traps E_2, E_3, H_1, and H_4, each assigned a unique $N_T = 10^{13}$ cm^{-3}, to assess

TABLE II. Changes in the PiN diode parameters after selectively excluding each trap from the TCAD simulation.

Excluded defect	I_F at 5 V	T_{rr} (ns)	I_{prr} (A)	Q_{rr} (μC)
E_1 ($E_C - 0.19$ eV)	55.3 A	75	−152	5.7
E_2 ($E_C - 0.67$ eV)	56.2 A	76	−153.6	5.9
E_3 ($E_C - 1.65$ eV)	57.9 A	80	−156.9	6.3
H_1 (Ev + 0.16 eV)	58.3 A	81	−163.3	6.5
H_2 (Ev + 0.30 eV)	55.3 A	74	−152	5.7
H_3 (Ev + 0.63 eV)	55.3 A	74	−152	5.7
H_4 (Ev + 1.43 eV)	57.5 A	77	−160.2	6.2

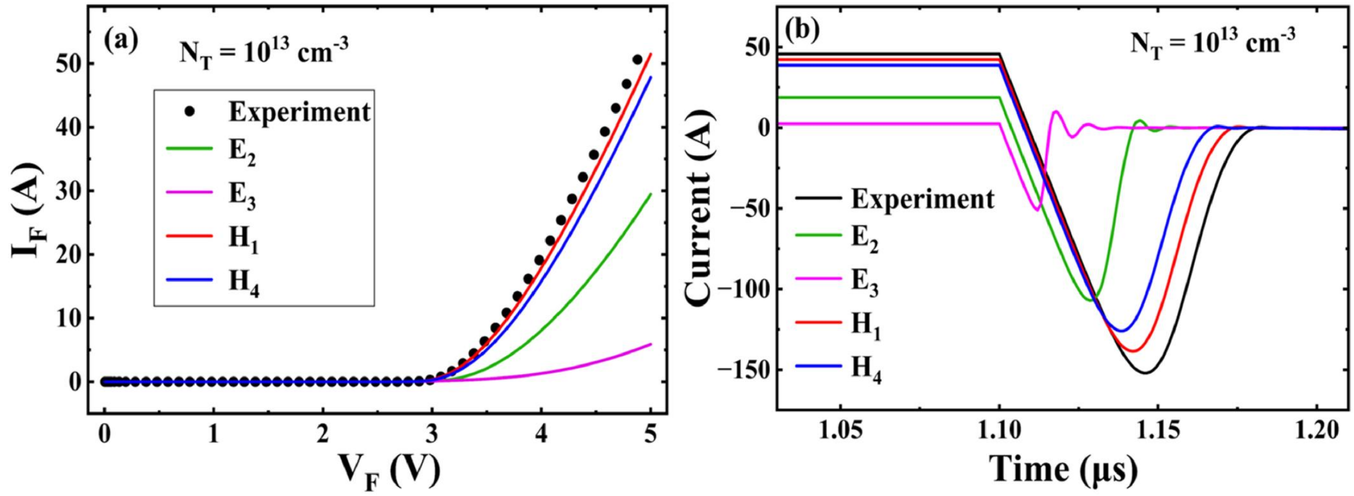

FIG. 13. Comparison of (a) I_F–V_F, and (b) RR simulation at a higher $N_T = 10^{13}$ cm^{-3} by selectively increasing N_T of traps E_2, E_3, H_1, and H_4 without changing other trap parameters in the TCAD model.

their relative impact on forward conduction. It is worth mentioning that the traps are selectively removed in earlier Secs. IV C and IV D, so the resultant I_F is increased for those cases. On the other hand, the N_T of a particular trap is increased in this section; consequently, I_F should decrease with N_T. In contrast to the pristine condition, H_1 shows a small deviation from the experimental curve, indicating minimal activity at higher N_T. The energy level of H_1 lies below the hole quasi-Fermi level E_{Fp}, as found from the simulated energy band diagram. Due to its energy location, H_1 may promote the hole emission by occupying an electron under high-injection conditions, with a minimal impact at higher N_T (compared to deep-level electron traps). As in the pristine case, traps E_1, H_2, and H_3 do not exert a notable influence on I_F–V_F properties at $N_T = 10^{13}$ cm^{-3}. Traps H_4 and E_2 cause moderate reductions in I_F, while E_3 leads to a substantial degradation. This pronounced effect stems from its deep donor level at $E_C - 1.65$ eV, which serves as an efficient SRH recombination center,[6] leading to a higher V_{ON} and lower I_F due to the increased series resistance.[22]

Figure 14(a) compares the simulated carrier concentration profiles along the drift layer depth at $V_F = 5$ V (high-injection condition), at a higher $N_T = 10^{13}$ cm^{-3} for E_2, E_3, and H_4. The midgap trap E_3 ($E_C - 1.65$ eV) is found to significantly suppress the

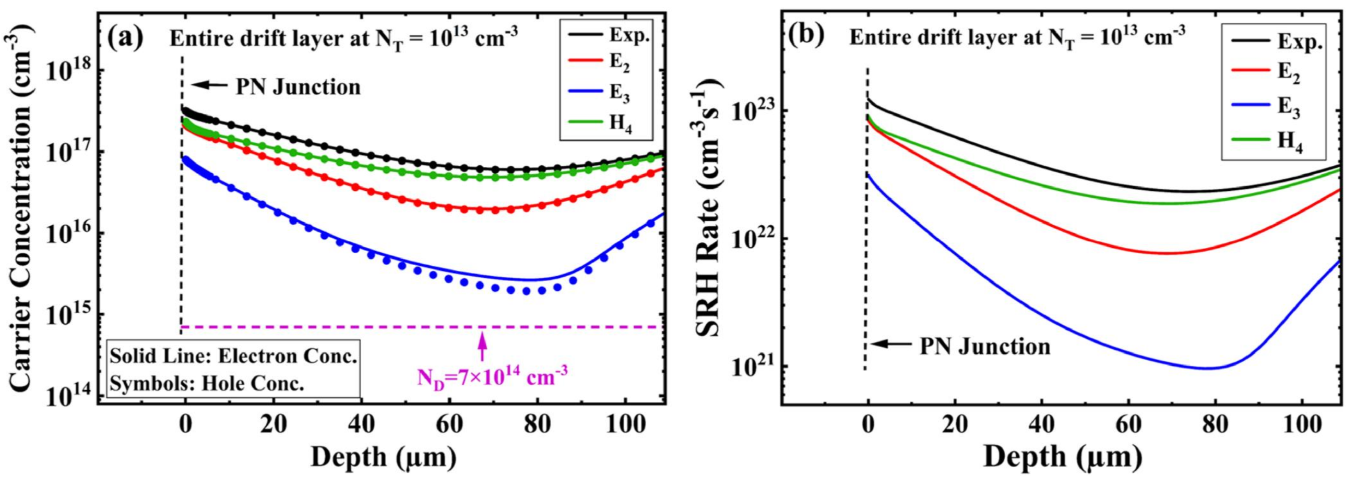

FIG. 14. (a) Carrier concentration profile and (b) SRH recombination rate, along the drift layer depth, are compared at a higher $N_T = 10^{13}$ cm^{-3} for traps E_2, E_3, and H_4 at $V_F = 5$ V.

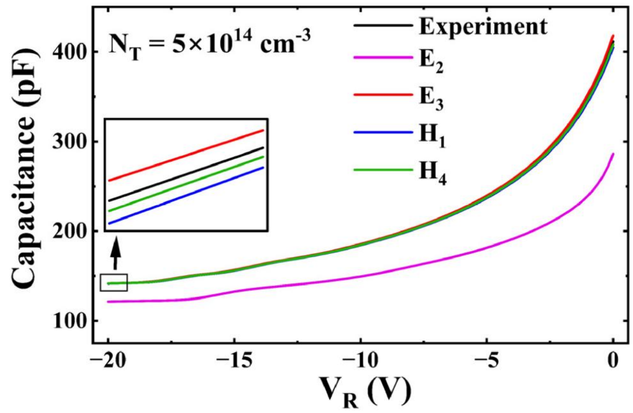

FIG. 15. Simulated C–V curve is compared at higher $N_T = 5 \times 10^{14}$ cm^{-3} by separately increasing N_T of each trap (E_2, E_3, H_1, and H_4) in the simulation model.

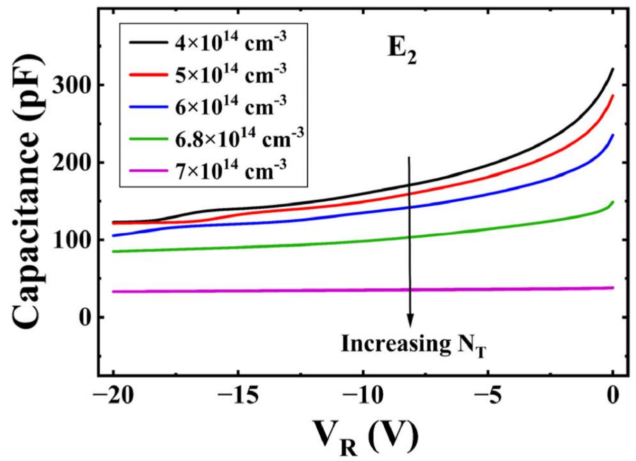

FIG. 16. Simulated C–V plot by only increasing the N_T (4×10^{14} to 7×10^{14} cm^{-3}) of the acceptor-type electron trap E_2.

injected carrier concentration in the drift layer, owing to its strong contribution to the SRH recombination. As a result, the carrier concentration is substantially decreased in Fig. 14(a) for the E_3 trap case (followed by E_2 and H_4), relative to the experimental conditions. Moreover, a notable difference in the injected electron and hole concentration is observed for the E_3 trap case (at $N_T = 10^{13}$ cm^{-3}), while the two curves (corresponding to E_2 and H_4) are almost identical. Since the trap E_3 reduces the injected carrier concentration, the SRH recombination rate is the lowest among all traps in Fig. 14(b). This observation is consistent with the carrier distribution profile in Fig. 14(a) and I_F–V_F characteristics in Fig. 13(a). The midgap trap E_3 reduces the reverse recovery time by accelerating carrier removal through efficient recombination of electrons and holes. Consequently, E_3 yields the shortest T_{rr} and lowest I_{prr} in Fig. 13(b), followed by E_2, H_4, and H_1, in agreement with forward current degradation.

Figure 15 shows the impact of individual traps on the C–V characteristics at $N_T = 5 \times 10^{14}$ cm^{-3}; this higher N_T value is chosen to clearly examine the acceptor-trapping-induced carrier compensation (CC) kinetics.[14,21] At this elevated N_T, E_2 significantly reduces the depletion capacitance via the CC effect. The deep-level acceptor E_2 compensates for the background donor doping density (N_D) in the n$^-$ drift layer at higher N_T values, thereby decreasing the net free carrier density and depletion capacitance (C), as realized in the below expression:[14,21]

$$C = A\sqrt{\frac{q\varepsilon_s\left(N_D^+ - N_T^-\right)}{2(V_R + V_{bi})}}, \tag{8}$$

where A is the diode area, N_D^+ is the positively ionized donor atoms in the depletion region, N_T^- is the negatively charged acceptor trap density, and V_{bi} is the built-in barrier potential of the PiN diode. Due to the acceptor-induced carrier compensation effect, diode depletion capacitance decreases with only increasing N_T of the trap E_2 in Fig. 16. Once the N_T of the trap E_2 reaches the background doping concentration (N_T of $E_2 \approx N_D$), the C–V curve becomes flat, as observed in Fig. 16 for $N_T = 7 \times 10^{14}$ cm^{-3}. This is because the donor doping density in the drift layer is completely compensated by the E_2 acceptor defects at $N_T = N_D$. Hence, the deep-level acceptor E_2 is primarily involved in the CC. On the other hand, a little increase in capacitance is seen for the E_3 case in the inset of Fig. 15 due to the positive space-charge contribution from donor trap E_3 to the effective doping density. The traps H_1 and H_4 have shown a barely minimal impact on the C–V characteristics, as perceived in Fig. 15.

The simulated I_F–V_F properties are evaluated at $N_T = 5 \times 10^{14}$ cm^{-3} (under drift-layer compensated condition) for each trap case in Fig. 17. The trapping-induced degradation in I_F is sequenced as follows: $E_2 > E_3 > H_4 > H_1 > H_3 > H_2 > E_1$; yet

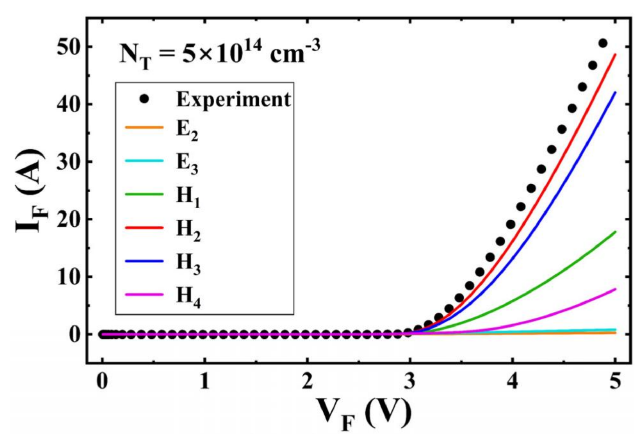

FIG. 17. Simulated I_F–V_F properties are evaluated at a higher $N_T = 5 \times 10^{14}$ cm^{-3} for each trap case E_2, E_3, H_1, H_2, H_3, and H_4.

TABLE III. Summary of electron and hole trapping impacts on 4H-SiC PiN power diode performance under pristine and higher N_T conditions.

Pristine diodes	I_F reduction	$H_1 > E_3 > H_4 > E_2$
	RR impact	$H_1 > H_4 > E_3 > E_2$
	CC effect	No change
High-N_T effect	I_F reduction	$E_3 > E_2 > H_4 > H_1$
	RR impact	$E_3 > E_2 > H_4 > H_1$
	CC effect	E_2

again, the midgap trap E_3 predominantly deteriorates the forward conduction. Table III summarizes the electron and hole trapping effects on I_F, RR, and CC properties of the 4H-SiC PiN power diodes, under pristine and higher N_T conditions. The H_1 exhibits a strong impact in reducing I_F (next only E_3) in pristine 4H-SiC PiN diodes. The *C–V* curve of the pristine diodes is unaffected by the traps, as their N_T is more than two orders lower than N_D. Whereas the deep-level electron traps E_2 and E_3 significantly reduce the diode conduction current at higher N_T. Among all traps, the deep acceptor E_2 is primarily responsible for the donor doping compensation in the n^- 4H-SiC drift layer.

V. CONCLUSION

The comprehensive electron and hole trapping investigations are performed in the 4H-SiC PiN power diodes using the DLTFS characterizations and TCAD simulations. The trap levels E_1, E_2, H_1, H_2, and H_3 identified from the DLTFS measurements are included in the TCAD simulation model, along with the two deep-level traps E_3 and H_4. The strong correlation between simulated and measured I_F–V_F, *C–V*, and *RR* characteristics confirms the accuracy of the TCAD physical models. The comparative studies reveal that hole traps exert a more pronounced influence on I_F and RR characteristics than electron traps due to their higher N_T and σ in the pristine PiN diodes. Among the trap levels, the shallow hole trap H_1 is identified as the dominant detrimental defect (next only to the midgap defect state E_3) in the pristine diodes. At higher trap concentrations, the midgap trap E_3 severely degrades I_F, followed by E_2, and H_4, as found from the simulation inspections. However, only the deep acceptor trap E_2 is essentially responsible (not the deep donor E_3) for the CC behavior in the 4H-SiC PiN diodes.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Shikha Kumari: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Besar Asllani:** Data curation (equal); Investigation (equal); Resources (equal). **Christophe Raynaud:** Investigation (equal); Resources (equal); Supervision (equal). **Dominique Planson:** Investigation (equal); Resources (equal); Supervision (equal). **Pierre Brosselard:** Investigation (equal); Resources (equal); Supervision (equal). **P. Vigneshwara Raja:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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